1. Summary

1.1 Backend & Database Work

I focused on improving the robustness and flexibility of the backend API, particularly around appointment management. I enhanced existing endpoints to support updates for scheduling, status changes, staff assignments, and notes. I also introduced safer request parsing and validation logic to reduce runtime errors and improve data consistency. Additionally, I investigated and resolved issues related to Prisma migrations in the production environment, particularly Enum conflicts during deployment.

1.2 Reminder System (Cron + Notifications)

I continued working on the appointment reminder system by testing the cron-based workflow responsible for sending notifications. This included validating time-window logic for multi-stage reminders (e.g., 5-day and 1-day reminders) and ensuring that duplicate notifications are prevented using database flags. I also verified that the system functions correctly in a local environment and began investigating inconsistencies in push notification delivery.

1.3 Calendar System: Cosmetic Changes

I made UI-level improvements to the calendar system to enhance usability and visual consistency. This involved refining component behavior, exploring alternative calendar implementations, and addressing minor type-related issues when integrating date inputs. The focus was on ensuring the calendar aligns well with the overall design system and provides a smoother user experience.

1.4 AI Integration

I continued developing the AI-powered features within the application, particularly the AI assistant. I structured the response schema to include actionable insights such as task prioritization, time planning, alerts, and risk evaluation. I also worked on integrating these outputs into the UI using Mantine (Our UI Framework) components, aiming to present AI-generated insights in a clear and user-friendly format.

1.5 Form Validations

I improved form handling across the application by strengthening validation logic to ensure accurate and consistent user input. This included refining data types, handling edge cases, and reducing the likelihood of invalid submissions. These improvements contribute to better data integrity and a more reliable user experience.

1.6 Architecture / State Management Reflection

I took time to reflect on the current state of the application's architecture, particularly around API handling and state management. I identified areas where my use of tools like TanStack Query and Zustand could be more structured and consistent. This reflection highlighted the need for better standardization in how data fetching, caching, and global state are managed, which I plan to improve moving forward.

2. Technical Decisions Taken

This week, I focused on improving the overall stability and structure of the application, particularly within the backend and data handling processes. I refined how appointment-related information is managed to ensure updates are handled more consistently and reliably. Alongside this, I addressed deployment-related issues to support smoother operation across different environments.

I also worked on testing the reminder system to verify that notifications are triggered at the appropriate times. This involved validating the timing logic and identifying areas where notification behavior could be more consistent. In parallel, I made several cosmetic improvements to the calendar interface to enhance usability and better align it with the overall design of the application.

Additionally, I continued integrating AI-driven features by improving how insights are generated and presented within the system, with a focus on making them more structured and user-friendly. I also strengthened form validation to ensure that user inputs are accurate and complete before submission, contributing to better data quality. Finally, I took time to reflect on the current architecture and state management approach, identifying areas where the system can be made more consistent and maintainable moving forward.

3. Overall Progress

Overall, the work completed during this period focused on improving the reliability, usability, and overall structure of the Eco Clean platform. Key efforts included enhancing backend stability, refining appointment handling workflows, and resolving deployment-related issues to ensure smoother operation across environments. Improvements were also made to form validation to support more accurate and consistent data entry throughout the system.

In addition, refinements were made to the scheduling interface, including improvements to the calendar system and UI consistency to better align with the application's design. Work was also carried out on integrating AI-driven features, particularly in developing and structuring an AI task assistant and improving how insights are presented within the platform.

Further progress included exploring integration with external AI services to support intelligent insights and enhancing the overall user experience through iterative UI improvements. Alongside implementation work, attention was given to reviewing application architecture, particularly around state management and data-fetching patterns, to improve maintainability and consistency. Collectively, these efforts strengthen both the technical foundation and user experience of the platform, supporting continued development and future feature expansion.

4. Work Logs

Date	Time	Description
2026-03-17	9:30–12:30pm	Worked on appointment API improvements, including handling scheduling updates, status changes, staff assignments, and notes
2026-03-17	1:30–2:30pm	Investigated Prisma migration failure and resolved enum conflict issue during Vercel deployment
2026-03-18	10:00–12:30pm	Refined calendar implementation, explored Mantine vs FullCalendar, and resolved date handling/type issues
2026-03-18	1:30–2:30pm	Adjusted UI styling to better match Mantine theme and improve visual consistency
2026-03-19	10:00–12:00pm	Worked on AI task assistant UI, structuring outputs and improving how insights are displayed
2026-03-19	11:00am–12:30pm	Attended project meeting to discuss progress, issues, and next steps

2026-03-20	1:00–2:00pm	Iterated on AI section design (gradients, layout, visual balance) for better dashboard integration
2026-03-20	10:00–11:00am	Attended project meeting to discuss progress, issues, and next steps
2026-03-20	11:30am–1:30pm	Explored OpenAI integration and planned how to incorporate AI insights into the application
2026-03-21	10:00–11:30am	Improved form validation and handled edge cases for user input consistency
2026-03-21	11:30am–1:00pm	Reviewed application architecture, focusing on API structure, TanStack Query usage, and Zustand state management
2026-03-21	1:30–2:30pm	Debugged client/server boundary issue related to data fetching and component usage in Next.js
Total	20.5h	

5. AI Usage

AI	Prompt	Value Addition
ChatGPT 5.3, Plus	Improving appointment API structure and handling updates for scheduling, status, and staff assignments	Helped refine backend logic for more consistent data handling and improved reliability of appointment updates
ChatGPT 5.3, Plus	Debugging Prisma migration failure related to enum conflicts during deployment	Identified root cause of migration error and guided resolution for successful deployment
ChatGPT 5.3, Plus	Designing and structuring AI task assistant response format for actionable insights	Provided structured schema for AI outputs including prioritization, time planning, and risk evaluation
ChatGPT 5.3, Plus	Improving UI design for AI insights section and refining visual balance and gradients	Suggested design improvements for cleaner, more user-friendly presentation of AI-generated content
ChatGPT 5.3, Plus	Connecting application to OpenAI API and planning integration of AI-driven insights	Guided integration approach and helped structure how AI features can be incorporated into the system

ChatGPT 5.3, Plus	Understanding proper usage of TanStack Query and Zustand for state management	Highlighted inconsistencies in current implementation and suggested more structured data flow patterns
ChatGPT 5.3, Plus	Resolving Next.js error related to using useQuery in a server component	Explained client vs server component boundaries and corrected data-fetching approach
ChatGPT 5.3, Plus	Improving form validation logic and handling edge cases in user input	Strengthened validation approach to improve data consistency and reduce input errors

6. Repo Check-in

Work on the repository was carried out on a regular basis, aligned with the project timeline and development milestones. Development activities included frontend UI components, backend API implementation, database schema evolution, PWA components and deployments.

Majority of the work for the eco-clean-web was committed through the branch **bug-fix-1, ai, stability and touchup**.

PR #13: https://github.com/vidarshan/W26_4495_S2_UpulA/pull/13
PR #14: https://github.com/vidarshan/W26_4495_S2_UpulA/pull/14
PR #15: https://github.com/vidarshan/W26_4495_S2_UpulA/pull/15
PR #16: https://github.com/vidarshan/W26_4495_S2_UpulA/pull/16
PR #17: https://github.com/vidarshan/W26_4495_S2_UpulA/pull/17
The work was checked into the repository almost every day, followed by a merge to main almost every day to publish changes to the live preview.